

Structural obstructions in coordinate basis selection

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Abstract

We give exact obstructions to three surrogate principles for selecting coordinate bases: maximizing products of principal minors, replacing projection noise by its convex hull, and selecting a basis by a Gaussian-target ℓ_1 optimization. Global principal-volume maximizers can have their best Nyström error separated from the volume mean by ratios tending to three. Even with constant precision diagonal, every constrained product maximizer in an explicit positive family has ratio tending to $(3 + \sqrt{6})/2$. A five-row Parseval example exhibits a strict gap between mixed noise and genuine rank-two projection noise; the gap survives an explicit perturbation making all coordinate bases nonsingular. A conditional Gaussian calculation yields a lower-tail estimate for the actual basis selected by ℓ_1 optimization, and a single deterministic frame sequence defeats every subharmonic cost threshold. Finally, an explicit Fourier reconstruction finds low-cost complete coordinate bases in a classical conference-square family. The conclusions concern the stated selection mechanisms and do not resolve unrestricted minimax basis selection.

Contribution and significance

The new content consists of exact counterexamples and estimates for specified coordinate-selection mechanisms. Global principal-volume maximizers can lose a factor approaching three; imposing constant precision diagonal still leaves the exact limit $(3 + \sqrt{6})/2$. A rational certificate separates mixed noise from genuine projection noise, and an explicit full-spark perturbation preserves the gap. For Gaussian-target ℓ_1 selection, a conditional law of the selected basis yields a tail obstruction that holds along a single deterministic sequence. Finally, a Fourier reconstruction supplies actual low-cost coordinate bases in a classical conference-square family.

The value lies in making the failures structural and quantifiable: global optimality rules out an optimization artifact, the full-spark perturbation rules out dependence on singular bases, and conditioning on the selected facet avoids treating a selected submatrix as independent Gaussian data. Classical determinant identities, projection convexity, basis pursuit and conference constructions are background. The contributions are the explicit certificates, exact limiting constants, selected-basis estimate and coordinate reconstruction; they concern the stated mechanisms rather than unrestricted minimax selection.

Relation to established tools. The projection hull is the classical Fantope used, for example, by Vu et al. [3]; basis pursuit is the established ℓ_1 principle of Chen, Donoho, and Saunders [4]. The conference construction is attributed to its Belevitch–Turyn lineage, as presented by Glazyrin [2]; our added result is an actual-coordinate reconstruction and inverse-cost bound for that family. The examples refute precisely specified surrogate guarantees, not previously published theorems. These distinctions locate the mathematical additions relative to the cited tools without asserting an exhaustive priority result.

1. Principal-volume optimization can favor the wrong orientation

For a positive definite real matrix K with spectrum λ , write

$$E_K(I) = \text{tr}(K - K_{:,I} K_{I,I}^{-1} K_{I,:}), \quad y_k(\lambda) = (k+1)e_{k+1}(\lambda)/e_k(\lambda), \quad \Phi_k(K) = \sum_{|I|=k} \log \det K_{I,I}.$$

The first two quantities arise in the fixed-spectrum comparison problem [1, Problem 4.7]. A global principal-volume maximizer means a maximizer of Φ_k over the entire real orthogonal orbit.

Lemma 1.1. *If a positive definite precision matrix T has every principal pair determinant equal to one, then $K = T^{-1}$ is a global maximizer of Φ_{n-2} for its spectrum. Moreover*

$$E_K(I) = \sum_{j \in I^c} T_{jj}, \quad y_{n-2}(\lambda) = 2 \text{tr } T/n.$$

Proof. Block Gaussian elimination gives the residual block $(T_{I^c, I^c})^{-1}$ and Jacobi's identity $\det K_{I,I} = \det K \det T_{I^c, I^c} = \det K$. Thus all $\binom{n}{2}$ selected principal determinants are equal. Their sum is the spectral invariant $e_{n-2}(\lambda)$, so concavity of the logarithm proves global product maximality. Inverse trace of a determinant-one 2×2 matrix equals its trace. The complementary one-minor identity gives $e_{n-1}(\lambda) = \det K \text{tr } T$, while $e_{n-2}(\lambda) = \binom{n}{2} \det K$. These imply the mean formula. $\square$

Theorem 1.2. *There are strictly positive spectra and global principal-volume maximizers K_r with*

$$\frac{y_r(\text{spec } K_r)}{\min_{|I|=r} E_{K_r}(I)} = \frac{3r+2}{r+2} \rightarrow 3.$$

Thus no factor $A < 3$ can guarantee $\min_I E_K(I) \geq y_k/A$ for every global principal-volume maximizer.

Proof. Set $n = r+2$ and let T_r have first two diagonal entries two, remaining diagonal entries six, entry $\sqrt{3}$ between the first two coordinates, entry $\sqrt{11}$ between either first coordinate and any remaining one, and entry $\sqrt{35}$ between distinct remaining coordinates. Its first block is $D = \begin{pmatrix} 2 & \sqrt{3} \\ \sqrt{3} & 2 \end{pmatrix} > 0$. The Schur complement of this block is

$$(6 - \sqrt{35})I_r + \left(\sqrt{35} - \frac{22}{2 + \sqrt{3}} \right) \mathbf{1}\mathbf{1}^T > 0.$$

The first coefficient is positive. The second is positive because $\sqrt{35}(2 + \sqrt{3}) > 22$, which follows successively from $140\sqrt{3} > 239$ and $58800 > 57121$. Hence $T_r > 0$. Every pair determinant equals one: the three types give $4 - 3$, $12 - 11$, and $36 - 35$. Lemma 1.1 applies. The smallest pair diagonal sum is four, and $\text{tr } T_r = 4 + 6r$. Thus the ratio is $(8 + 12r)/(4(r+2))$, as stated. $\square$

A four-dimensional example already violates factor two. Take a precision matrix with diagonal

$$(5/3, 5/3, 101/20, 101/20),$$

within-pair entries $4/3, 99/20$, and every cross entry $\sqrt{267}/6$. Every pair determinant is one. The two antisymmetric eigenvalues are $1/3, 1/10$, and the symmetric-pair block is

$$\begin{pmatrix} 3 & \sqrt{267}/3 \\ \sqrt{267}/3 & 10 \end{pmatrix},$$

with positive determinant $1/3$. Its inverse has eigenvalues $(39 \pm \sqrt{1509})/2$. Thus the covariance spectrum is $(3, 10, (39 + \sqrt{1509})/2, (39 - \sqrt{1509})/2)$, all positive. The minimum error is $10/3$, whereas $y_2 = 403/60$, so $y_2/2 - \min E = 1/40$.

2. Constant precision diagonal does not repair product selection

Theorem 2.1. For $0 < \varepsilon < 1/4$ put $\mu(\varepsilon) = (4 - \varepsilon - 2\varepsilon^2, \varepsilon, \varepsilon^2, \varepsilon^2)$. Let $\mathcal{B}_\varepsilon$ be the real symmetric matrices with that spectrum and diagonal identically one, and let $\mathcal{M}_\varepsilon$ maximize $\sum_{i < j} \log \det T_{\{i,j\}}$ on this set. Both sets are nonempty and compact. If

$$f(T) = \min_{i < j} \text{tr}(T_{\{i,j\}}^{-1}), \quad Z(\mu) = 12/e_2(\mu),$$

then

$$\lim_{\varepsilon \downarrow 0} \sup_{T \in \mathcal{M}_\varepsilon} \left| \frac{Z(\mu(\varepsilon))}{f(T)} - \frac{3 + \sqrt{6}}{2} \right| = 0.$$

In particular every constrained global maximizer violates the factor-two comparison for all sufficiently small positive ε .

Proof. Put $\delta = \varepsilon^2$, $r = 1 - \delta$, $b = \varepsilon - \varepsilon^2$, $a = 4 - \varepsilon - 3\varepsilon^2$; then $a + b = 4r$ and $r - b = 1 - \varepsilon > 0$. For a unit vector $v \in \mathbb{R}^4$ define $w_i(v) = \sqrt{r - bv_i^2}$ and

$$\mathcal{F}_\varepsilon = \{v : \|v\| = 1, \sum_i v_i w_i(v) = 0\}.$$

For such a vector, $u_i = w_i/\sqrt{a}$ is unit and orthogonal to v , so $T_\varepsilon(v) = \delta I + auu^T + bv v^T$ has the stated spectrum and diagonal one. Conversely any member of $\mathcal{B}_\varepsilon$ has this form up to coordinate sign changes: in its spectral decomposition the diagonal condition gives $au_i^2 = r - bv_i^2 > 0$, so sign changes make each u_i positive. These changes preserve every principal determinant.

Define

$$\alpha = \sqrt{(3 + \sqrt{6})/12}, \quad \beta = \sqrt{(3 - \sqrt{6})/12}, \quad v^* = (-\alpha, -\beta, \beta, \alpha).$$

It lies in every $\mathcal{F}_\varepsilon$ by cancellation of opposite coordinate pairs, proving nonemptiness. Compactness of the orthogonal orbit, the closed diagonal constraint, and positivity of all pair determinants prove existence of maximizers.

The exact pair determinants are

$$D_{ij}(T_\varepsilon(v)) = 2\delta - \delta^2 + b(w_i v_j - w_j v_i)^2.$$

Put $q_\varepsilon = (2\delta - \delta^2)/b$, $L_{ij,\varepsilon} = q_\varepsilon + (w_i v_j - w_j v_i)^2$, $P_\varepsilon = \prod_{i < j} L_{ij,\varepsilon}$, and $H_\varepsilon = \max_{i < j} L_{ij,\varepsilon}$. Product maximization is equivalent to maximizing P_ε on $\mathcal{F}_\varepsilon$. Since $|w_i - 1| \leq \varepsilon$, these functions converge uniformly on the unit sphere to

$$P_0(v) = \prod_{i < j} (v_i - v_j)^2, \quad H_0(v) = \max_{i < j} (v_i - v_j)^2.$$

Every limit of feasible v as $\varepsilon \rightarrow 0$ belongs to $\mathcal{S}_0 = \{v : \sum v_i = 0, \|v\| = 1\}$, because $|\sum v_i| \leq 2\varepsilon$.

We solve the limiting problem completely. Its maximum is positive, so a maximizer x has distinct coordinates. Lagrange multipliers for $\log P_0$ give $2 \sum_{j \neq i} (x_i - x_j)^{-1} = 2\ell x_i + h$. Summation gives $h = 0$; multiplication by x_i and summation gives $\ell = 6$. Let $p(z) = \prod_i (z - x_i) = z^4 - z^2/2 + cz + t$. Then $p''(x_i) = 12x_i p'(x_i)$ for every root. Thus $p'' - 12zp' + 48p$, a polynomial of degree at most three vanishing at four distinct roots, is zero. Its expansion is $36cz + 48t - 1$, whence $c = 0, t = 1/48$. Its roots are exactly the entries of v^* . Every limiting maximizer is a permutation of v^* and has $H_0 = (3 + \sqrt{6})/3$.

For any sequence of true constrained maximizers, represent them by maximizing $v_\varepsilon \in \mathcal{F}_\varepsilon$ and pass to a convergent subsequence. The exactly feasible comparison v^* gives $P_\varepsilon(v_\varepsilon) \geq P_\varepsilon(v^*)$.

Uniform convergence forces its limit to maximize P_0 on S_0 . Hence $\max D_{ij}/b \rightarrow (3 + \sqrt{6})/3$ uniformly over the maximizing set; failure of uniformity would supply a contrary sequence and then a convergent subsequence of precisely this kind. Finally $f(T) = 2/\max D_{ij}$, and

$$e_2(\mu) = 4\varepsilon + 7\varepsilon^2 - 2\varepsilon^3 - 3\varepsilon^4, \quad e_2(\mu)/b \rightarrow 4.$$

Therefore $Z/f = 6 \max D_{ij}/e_2(\mu) \rightarrow (3 + \sqrt{6})/2$ uniformly. Complementary-minor and Schur-complement identities identify Z and f with the covariance volume mean and best error. The limit exceeds two, proving the last claim. $\square$

3. Mixed noise is strictly stronger than projection noise

For $U \in \mathbb{R}^{n \times k}$ with $U^T U = I_k$, write $P = I - U U^T$, and let $\mathcal{B}(U)$ contain all nonsingular k -row coordinate bases. With coordinate embedding E_I , set

$$A_I = P E_I (U_I U_I^T)^{-1} E_I^T P, \quad \mathcal{F}_q(P) = \{X = X^T : 0 \leq X \leq P, \text{tr } X = q\},$$

where $1 \leq q \leq n - k$. Define

$$R_q(U) = q + \max_{X \in \mathcal{F}_q(P)} \min_{I \in \mathcal{B}(U)} \text{tr}(X A_I),$$

$$H_q(U) = \max_{V^T V = I_q, U^T V = 0} \min_{I \in \mathcal{B}(U)} [q + \|U_I^{-1} V_I\|_F^2].$$

Proposition 3.1. *The set $\mathcal{F}_q(P)$ is the convex hull of rank- q orthogonal projections supported on range P . Moreover*

$$R_q(U) = q + \min_{p \in \Delta(\mathcal{B}(U))} \sum_{j=1}^q \lambda_j^\downarrow \left(\sum_I p_I A_I \Big|_{\text{range } P} \right), \quad H_q(U) \leq R_q(U) \leq q(k+1).$$

Proof. If $0 \leq X \leq P$, then X vanishes on $\ker P$, and its eigenvalues on $\text{range } P$ lie in $[0, 1]$ and sum to q . Moving mass between two fractional entries until one reaches zero or one, in both directions, expresses this eigenvalue vector recursively as a convex combination of zero-one vectors of sum q . Keeping the eigenbasis proves the hull statement. Genuine noise has $X = V V^T$ and cost $\text{tr}(X A_I)$.

For completeness, let $\mathcal{K}$ be the compact convex image of $\mathcal{F}_q(P)$ under $X \mapsto (\text{tr}(X A_I))_I$, and put $r = \max_{z \in \mathcal{K}} \min_I z_I$. For $t > r$, separate $\mathcal{K}$ from the orthant $\{z : z_I \geq t\}$. A closest pair exists by compactness and coercivity of distance. The vector between this pair separates the sets by the first variation of squared distance, has nonnegative coordinates since the orthant is upward-closed, and can be normalized to a probability vector p . It gives $\max_{z \in \mathcal{K}} p \cdot z < t$. Taking $t \downarrow r$, together with the reverse elementary inequality, proves finite-dimensional minimax. For fixed p , diagonalizing $\sum p_I A_I$ shows that maximization over X fills its largest q eigenvalues, giving the formula.

Use the probability weights $p_I = \det(U_I)^2$. For $w \in \text{range } P$, Cramer's rule expresses each coordinate of $U_I^{-1} w_I$ as a replaced-column minor divided by $\det U_I$. Cauchy–Binet over all subsets gives squared-minor sum $\|w\|^2$ for each replaced top column; omitting singular original bases only decreases the sum. Therefore $\sum_I p_I w^T A_I w \leq k \|w\|^2$, or $\sum p_I A_I \leq k P$. The top- q sum is at most kq . $\square$

Theorem 3.2. *Let U_0 have rows*

$$u_1 = u_2 = (1/2, 1/\sqrt{10}), \quad u_3 = u_4 = (-1/2, 1/\sqrt{10}), \quad u_5 = (0, -\sqrt{3/5}).$$

Then $U_0^T U_0 = I_2$ and $R_2(U_0) = 100/19$, while $H_2(U_0) \leq 5$. Let $s = 10^{-4}$, $c = \sqrt{1 - s^2}$, and let O_s rotate coordinates 1, 3 by $\begin{pmatrix} c & s \\ -s & c \end{pmatrix}$ and fix the other coordinates. For $U_s = O_s U_0$, all ten row pairs are nonsingular and

$$R_2(U_s) > 21/4 > 3 + \sqrt{5} > 51/10 > H_2(U_s).$$

Proof. An orthonormal complement of U_0 consists of

$$\begin{aligned} c_1 &= (1, -1, 0, 0, 0)^T / \sqrt{2}, & c_2 &= (0, 0, 1, -1, 0)^T / \sqrt{2}, \\ v &= (\sqrt{3/20}, \sqrt{3/20}, \sqrt{3/20}, \sqrt{3/20}, \sqrt{2/5})^T. \end{aligned}$$

Put $P_0 = c_1 c_1^T + c_2 c_2^T + v v^T$ and $X_0 = (10/19)P_0 + (8/19)v v^T$. Its complement eigenvalues are 10/19, 10/19, 18/19, so it is feasible mixed noise of trace two. The eight nonsingular pairs split into $\mathcal{A} = \{13, 14, 23, 24\}$ and $\mathcal{B} = \{15, 25, 35, 45\}$. If $h_I = \text{tr}(U_I U_I^T)^{-1}$ and $z_I = v_I^T (U_I U_I^T)^{-1} v_I$, direct inversion gives

$$(h_I, z_I) = (7, 3/2) \quad (I \in \mathcal{A}), \quad (h_I, z_I) = (19/3, 7/3) \quad (I \in \mathcal{B}).$$

For clarity the first row Gram is $\begin{pmatrix} 7/20 & -3/20 \\ -3/20 & 7/20 \end{pmatrix}$ with inverse $\begin{pmatrix} 7/2 & 3/2 \\ 3/2 & 7/2 \end{pmatrix}$. The second has diagonal 7/20, 3/5, off-diagonal $-\sqrt{6}/10$, and determinant 3/20. As $\text{tr } A_I = h_I - 2$, every mixed cost is

$$2 + \text{tr}(X_0 A_I) = \frac{18 + 10h_I + 8z_I}{19} = \frac{100}{19}.$$

For a dual certificate give each $\mathcal{A}$ pair weight 4/19 and each $\mathcal{B}$ pair weight 3/76. In the complement basis (c_1, c_2, v) , uniform averaging within the classes cancels off-diagonals by the swaps $1 \leftrightarrow 2$, $3 \leftrightarrow 4$. The class-average diagonals are respectively $(7/4, 7/4, 3/2)$ and $(1, 1, 7/3)$. Their weighted average is $(31/19)I_3$. Thus every feasible X has averaged cost $2 + (31/19) \text{tr } X = 100/19$, proving the exact relaxed value.

Genuine rank-two noise is $P_0 - w w^T$ with $w = \alpha c_1 + \beta c_2 + \gamma v$ unit. Put $a = \max(|\alpha|, |\beta|)$, $b = \min(|\alpha|, |\beta|)$, $g = |\gamma|$. Maximizing the subtracted quadratic form by choosing the signs of contrast rows shows that the minimum genuine costs in the two classes are

$$\begin{aligned} A &= 21/4 + g^2/4 - (3/2)ab - \sqrt{15/2}g(a+b), \\ B &= 19/3 - 2a^2 - (7/3)g^2 - 2\sqrt{10/3}ag. \end{aligned}$$

These formulas follow by multiplying by the two displayed inverse Grams; all signs can be made favorable simultaneously within each class. If $B \leq 5$, the desired bound holds. Otherwise $2a^2 + (7/3)g^2 < 4/3$. Since $a^2 \geq (1 - g^2)/2$, this gives $g^2 < 1/4$, and $b^2 = 1 - g^2 - a^2 > 1/3 + g^2/6$. Hence $ab \geq b^2 > 1/3$, and $A < 21/4 + 1/16 - 1/2 = 77/16 < 5$. Thus $H_2(U_0) \leq 5$.

For the perturbation, the first four rows are $(a_i/2, b_i/\sqrt{10})$, with

$$(a_1, b_1) = (c - s, c + s), \quad (a_2, b_2) = (1, 1), \quad (a_3, b_3) = (-c - s, c - s), \quad (a_4, b_4) = (-1, 1).$$

Their unscaled pair determinants are $-2s, -2s, 2, 2, 2c, 2c$ for pairs 12, 34, 13, 24, 14, 23 respectively. Pairs with row five are nonsingular because $a_i \neq 0$. Conjugate X_0 by O_s to obtain feasible X_s . Write $d = cs$. For first-four pairs,

$$\begin{aligned} h_{13} &= h_{24} = 7, & h_{14} &= (7 - 3d)/c^2, & h_{23} &= (7 + 3d)/c^2, \\ h_{12} &= (7 - 3d)/s^2, & h_{34} &= (7 + 3d)/s^2, \end{aligned}$$

and $z_I = 3/2$. Their mixed costs are $(30 + 10h_I)/19$. Pairs 25, 45 retain cost 100/19; pairs 15, 35 have costs

$$\frac{205 - 2d}{191 - 2d}, \quad \frac{205 + 2d}{191 + 2d}.$$

For the latter formulas one may use $h_{i5} = 5/3 + (2/3)b_i^2/a_i^2 + 4/a_i^2$ and $z_{i5} = 2/3 + (5/3)b_i^2/a_i^2$ in $(18 + 10h + 8z)/19$. Since $0 < d < s < 1/1000$, every first-four cost exceeds $9997/1900 > 21/4$; the 15 cost exceeds 100/19, while the 35 cost exceeds $2496/475 > 21/4$. Thus $R_2(U_s) > 21/4$.

For a uniform genuine-noise comparison use $w_s = O_s w$. On each original nonsingular pair the Gram G has smallest eigenvalue at least $3/20$. Since $\|O_s - I\|_2 \leq 2s$,

$$\|G_s - G\|_2 \leq 4s, \quad \|(w_s)_I - w_I\| \leq 2s,$$

and both inverse Grams have norm at most seven because $3/20 - 4s > 1/7$. The inverse identity gives $\|G_s^{-1} - G^{-1}\|_2 \leq 196s$, so inverse traces differ by at most $392s$. The subtracted quadratic forms differ by at most $196s + 28s = 224s$. For every w , a formerly available pair of cost at most five therefore still has cost at most $5 + 616s < 51/10$. Taking the minimum over all ten pairs and then the maximum proves the upper bound on $H_2(U_s)$. The central comparisons follow from $\sqrt{5} < 9/4$ and $\sqrt{5} > 21/10$. $\square$

Proposition 3.3 (The four-row comparison core). *For real Parseval matrices $Q \in \mathbb{R}^{4 \times 2}$,*

$$\sup_Q \min_{|I|=2, \det Q_I \neq 0} \|Q_I^{-1}\|_F^2 = 3 + \sqrt{5}.$$

Thus the relaxed value in Theorem 3.2 exceeds the corresponding genuine four-row core.

Proof. Put $p_{ij} = \det Q_{\{i,j\}}$, $w_{ij} = p_{ij}^2$. Then $\sum w_{ij} = 1$, and expansion using column orthonormality gives $\sum_{j \neq i} w_{ij} = \|q_i\|^2$. Hence for a nonsingular pair I ,

$$\|Q_I^{-1}\|_F^2 = 1 + \frac{1 - w_{I^c}}{w_I}.$$

Let $t = 2 + \sqrt{5}$, $\alpha = 1/t$, $\beta = 2/(t+1)$, so $\alpha + 2\beta = 1$. If every eligible cost exceeded $t+1$, each complementary weight pair (a, b) would satisfy $ta + b < 1$, $a + tb < 1$. These remain true when one weight vanishes, using the other inequality. Their sums s_i lie strictly between α and β . With $z_i = \sqrt{ab}$, the inequalities give

$$z_i \geq f(s_i), \quad f(s) = \frac{\sqrt{(ts-1)(1-s)}}{t-1}.$$

The function is concave on $[\alpha, \beta]$, by concavity and nonnegativity of its quadratic argument and concavity and monotonicity of square root. Its endpoint values are $0, \beta/2$. Thus for distinct i, j, ℓ ,

$$z_j + z_\ell \geq \frac{\beta(1-s_i-2\alpha)}{2(\beta-\alpha)} > \frac{\beta}{2} > \frac{s_i}{2} \geq z_i.$$

But the real Plücker identity says the three signed quantities with absolute values z_i sum to zero, so they cannot satisfy all strict triangle inequalities. This proves the upper bound. For attainment put $a = \sqrt{5} - 2$ and take rows

$$(\sqrt{(1-a)/2}, 0), \quad (\sqrt{(1-a)/2}, 0), \quad (-\sqrt{a/2}, 1/\sqrt{2}), \quad (-\sqrt{a/2}, -1/\sqrt{2}).$$

The matrix is Parseval. The first pair is singular, each cross pair has inverse trace $4/(1-a) = 3 + \sqrt{5}$, and the last has trace $1 + 1/a = 3 + \sqrt{5}$. $\square$

In the core dimension $n = k + q$, complete $[U \ V]$ to the square orthogonal matrix itself. Complementary coordinate blocks satisfy $U_I U_I^T = I - V_I V_I^T$ and $V_J^T V_J = I - V_I^T V_I$. Their common nonunit eigenvalues show singularity is simultaneous, and otherwise

$$q + \|U_I^{-1} V_I\|_F^2 = \|V_J^{-1}\|_F^2.$$

Hence the supremum genuine-noise value in that dimension is the Parseval q -basis extremum. The explicit five-row example shows that its proposed extension to mixed noise fails. Since its genuine value remains smaller, it does not disprove the analogous extension for genuine projection noise.

4. The actual Gaussian ℓ_1 -selected basis

Fix a finite enumeration of all square coordinate bases. Given a full-rank design and a target, solve each nonsingular square system and select a representation minimizing its ℓ_1 norm, breaking ties by that enumeration. This attains the true ℓ_1 optimum: a minimum with dependent nonzero support can move along a support dependence, with zero objective slope, until a coordinate vanishes; iteration yields independent support, which extends to a square basis. The finite rule is measurable and is unchanged by simultaneous invertible feature transformations or positive rescaling of the target.

General-position convention. A $k \times n$ design is in *full general position* if every k columns are linearly independent and every $k + 1$ signed columns with distinct underlying column indices are affinely independent. A row frame has this property when its transpose does. These conditions imply that the symmetric column hull has simplex facets. A target is additionally required to avoid the spans of fewer than k columns when uniqueness and full selected support are used; that exceptional target set has Lebesgue measure zero. Both genericity conditions are preserved by invertible feature transformations.

Theorem 4.1. *Let $k \geq 4$ and let $X \in \mathbb{R}^{k \times 2k}$ have independent standard Gaussian entries. Fix a deterministic nonzero target. Its ℓ_1 optimal representation is almost surely unique and uses k nonzero coefficients. If J is the complete selected basis, then for every $t > 0$,*

$$\mathbb{P} \left\{ \|X_J^{-1}\|_F^2 \leq \frac{k-2}{2t} \right\} \leq 2(1+t/2)e^{-t/2}.$$

Proof. Orthogonal invariance and target rescaling reduce to target e_1 . Almost surely every k columns are independent, no $k + 1$ signed distinct columns are affinely dependent, and e_1 is in no span of $k - 1$ columns. Each exception is a finite union of zero sets of nonzero determinant polynomials. The symmetric hull of the columns has simplex facets, and the ray through e_1 meets a facet interior: otherwise the target would belong to a span of at most $k - 1$ columns. Its radial gauge describes the ℓ_1 optimum, proving uniqueness and full support.

Fix a set of k columns and signs, and call the signed matrix A . It is selected exactly when

$$u = A^{-1}e_1 > 0, \quad |g^T v| < 1 \quad \text{for every outside column } g, \quad v = A^{-T}\mathbf{1}.$$

Necessity follows from the selected supporting facet; sufficiency follows from $v^T A = \mathbf{1}^T$, $v^T e_1 = \mathbf{1}^T u$, and the dual inequalities. These events partition a probability-one set.

Put $r = \|u\|$, $n = u/r$, and choose a locally smooth orthonormal complement W to n . In domain coordinates $[n, W]$ and range coordinates $[e_1, e_1^\perp]$,

$$A = \begin{pmatrix} 1/r & a^T \\ 0 & D \end{pmatrix}, \quad D \in \mathbb{R}^{(k-1) \times (k-1)}.$$

The change of variables from A to (u, a, D) has Jacobian $|\det D|r^{-(k+1)}$. Indeed differentiating $Au = e_1$ gives $(dA)n = -A du/r$. Its determinant from du to the first column component is $|\det A|/r^k = |\det D|/r^{k+1}$; moving-complement terms affect only the remaining columns, so the full differential is block triangular. Patches cover the positive orthant and suffice for density integration.

Let $\alpha = \mathbf{1}^T u > 0$, $b = W^T \mathbf{1}$, and write $v = (\alpha, w)$. The remaining dual equations give $a = (b - D^T w)/\alpha$. The Jacobian from w to a at fixed u, D is $|\det D|/\alpha^{k-1}$. Conditional on u, w and this signed selection event, the density of D is therefore proportional to

$$(\det D)^2 \exp \left[-\frac{1}{2} \left(\|D\|_F^2 + \frac{\|b - D^T w\|^2}{\alpha^2} \right) \right]. \quad (4.1)$$

No unrecorded selection factor depends on D : the probability of all outside dual inequalities is $[\mathbb{P}\{|Z| < (\alpha^2 + \|w\|^2)^{-1/2}\}]^k$, depending only on u, w .

Rotate the rows of D to put w in the first direction. Write its first row as d and the remaining $p = k - 2$ rows as $G \in \mathbb{R}^{p \times (k-1)}$. Before the determinant tilt, completing the square in (4.1) gives standard independent Gaussian G and an independent row $d \sim N(\mu, \sigma^2 I)$, with

$$\sigma^2 = \frac{\alpha^2}{\alpha^2 + \|w\|^2}, \quad \mu = \frac{\|w\|b}{\alpha^2 + \|w\|^2}.$$

If η spans the kernel line of G , then $(\det D)^2 = \det(GG^T)(d \cdot \eta)^2$. Integrating d weights the Gaussian law of G by $\det(GG^T)[\sigma^2 + (\mu \cdot \eta)^2]$. The Gram $H = GG^T$ is independent of the orthonormal row frame $H^{-1/2}G$: conditional orthogonal invariance acts transitively on the latter, so its conditional law is the unique invariant probability and does not depend on H . The angular factor therefore integrates to a constant, leaving precisely a $\det H$ tilt of the usual Gram law.

This tilted Gram is the Gram of a $p \times (p + 3)$ standard Gaussian matrix. To see this directly, row Gram–Schmidt gives a lower triangular factor with independent standard normal subdiagonal entries and independent chi diagonal entries of degrees $m, m - 1, \dots, m - p + 1$, where $m = p + 1$. Multiplication by $\det H$ multiplies each chi density by its squared argument, raising each degree by two, precisely the factorization for $m + 2 = p + 3$ columns. For this Gram, each diagonal entry of H^{-1} has law $1/V$ with $V \sim \chi_4^2$: condition on the other $p - 1$ rows and take the squared Gaussian length in their four-dimensional orthogonal complement. No independence between inverse diagonal entries is claimed. If $\text{tr } H^{-1} \leq p/(2t)$, at least half the diagonal entries are at most $1/t$. Selecting a uniform independent diagonal index and averaging gives

$$\mathbb{P}\{\text{tr } H^{-1} \leq p/(2t)\} \leq 2\mathbb{P}\{V \geq t\} = 2(1 + t/2)e^{-t/2}.$$

Finally $\|A^{-1}\|_F^2 \geq \|D^{-1}\|_F^2 \geq \text{tr } H^{-1}$, the last inequality following by the principal inverse Schur complement. The bound holds conditionally on every u, w , then on each signed event, and hence for the actually selected basis after integration and summation. $\square$

5. One frame sequence defeats every subharmonic threshold

Put $B_k = \text{diag}(1, 1/2, \dots, 1/k)$ and $H_k = \sum_{j=1}^k 1/j$. For a $2k \times k$ Parseval frame R , use the preceding fixed ℓ_1 rule for $R^T x = y$ and write

$$M_R(y) = (R_{J_R(y)}^T R_{J_R(y)})^{-1}, \quad C_R(y) = \text{tr}(B_k M_R(y)).$$

Theorem 5.1. *There is a deterministic sequence of Parseval frames in full general position R_k , for $H_k \geq 6$, such that simultaneously for every $s \geq 0$,*

$$\mathbb{P}_{z \sim N(0, I_k)}\{C_{R_k}(B_k^{1/2} z) \leq s\} \leq \frac{L(s + 2k)}{kH_k}, \quad L = 320000000000.$$

Consequently the probability tends to zero for every deterministic sequence $s_k = o(kH_k)$, including $s_k = Ak$ for every fixed A .

Proof. Take X from Theorem 4.1, $S = XX^T$, and $R = X^T S^{-1/2}$. Then R is an invariantly distributed Parseval frame independent of S : right orthogonal invariance of X fixes S and acts transitively on the Stiefel frame, so the conditional invariant law is independent of S . Let $c = 10^{-8}$. We first establish

$$\mathbb{P}\{\lambda_{\min}(S) < ck\} \leq 3e^{-4k}. \quad (5.1)$$

A maximal ε -separated set on a d -sphere is an ε -net of cardinality at most $(1 + 2/\varepsilon)^d$, by the disjoint-ball volume comparison. Two $1/4$ -nets and a Gaussian tail bound give $\mathbb{P}\{\|X\|_2 > 10\sqrt{k}\} \leq 2 \cdot 9^{3k} e^{-25k/2} \leq 2e^{-5k}$. Set $a = 10^{-4}$. If the norm bound holds and $\|X^T v\| < a\sqrt{k}$ for some unit v , an $a/10$ -net has a point v' with $\|X^T v'\| < 2a\sqrt{k}$. The exponential moment bound for a d -dimensional Gaussian is $\mathbb{P}\{\|g\|^2 \leq b\} \leq (eb/d)^{d/2}$, $0 < b < d$, obtained by optimizing Markov's inequality for $e^{-\theta\|g\|^2}$. The union probability is at most $[(1 + 20/a)2ea^2]^k \leq e^{-4k}$; here $e < 3$ makes the bracket less than $0.012001 < 1/81 < e^{-4}$. Combining the bounds proves (5.1).

For any Parseval frame $M_R(y) \geq I$, hence $T_R(y) := \text{tr } M_R(y) \geq k$. The Gaussian problem at e_1 is equivalent to the Parseval target $S^{-1/2}e_1$, with the same selected indices. On the good event in (5.1), its trace is at least $ck\|X_J^{-1}\|_F^2$. The Haar law of $T_R(y)$ is independent of the nonzero target direction by right orthogonal invariance, and rescaling does not change the selected basis. Conditional on S , its target has this same law. Theorem 4.1, $k - 2 \geq k/2$, and a union bound therefore imply, for a trace T with this law,

$$\mathbb{P}\{T \leq ck^2/(4t)\} \leq f(t) + 3e^{-4k}, \quad f(t) = 2(1 + t/2)e^{-t/2}.$$

Integrating the lower tail, using $T \geq k$ and $\int_0^\infty f(t) dt = 8$, gives

$$\mathbb{E}(1/T) \leq 33/(ck^2). \quad (5.2)$$

Indeed integrate $\mathbb{P}(T < 1/u)$ over $0 < u < 1/k$ and substitute $t = ck^2u/4$; the exponentially small remainder is at most $1/(ck^2)$.

Fix a unit target e . The feasible representation Re has ℓ_1 norm at most $\sqrt{2k}$, so optimality gives $e^T M_R(e)e \leq 2k$. Let N be the compression of $M_R(e)$ to $e^\perp$ and $V = \text{tr } N$. Then $V \geq k - 1$ and $V \geq T - 2k$. Thus

$$\mathbb{E}(1/V) \leq 2\mathbb{E}(1/T) + \frac{\mathbb{P}(T < 4k)}{k - 1} \leq 400/(ck^2). \quad (5.3)$$

For the explicit constant use the last tail bound at $t = ck/16$ and put $x = ck/32$. Multiplication of the second term by ck^2 , using $1/(k - 1) \leq 2/k$, bounds it by $128x(1 + x)e^{-x} + 6cke^{-4k} \leq 257$; use $xe^{-x} \leq 1$, $x^2e^{-x} \leq 1$. The first term contributes at most 66.

We need an angular reciprocal estimate. For positive definite $D \in \mathbb{R}^{d \times d}$, $d \geq 3$, write $h = \text{tr } D$, $b = \|D\|_2$, and assume $h > 2b$. If q is uniform on the unit sphere, then

$$\mathbb{E} \frac{1}{q^T D q} \leq \frac{d - 2}{h - 2b}. \quad (5.4)$$

For a Gaussian $g = rq$, radius and direction are independent and $\mathbb{E}r^{-2} = 1/(d - 2)$. Tonelli and one-dimensional Gaussian integration give $\mathbb{E}(g^T D g)^{-1} = \int_0^\infty \prod_i (1 + 2t\beta_i)^{-1/2} dt$. Concavity of $\log(1 + 2tx)$ on $[0, b]$ bounds the integrand above by $(1 + 2tb)^{-h/(2b)}$, whose integral is $1/(h - 2b)$. This proves (5.4). If Q is Haar orthogonal and $N > 0$ has trace v , convexity of reciprocal applied to the eigenvalue weights of N gives

$$\mathbb{E}_Q \frac{1}{\text{tr}(DQNQ^T)} \leq \frac{d - 2}{(h - 2b) \text{tr } N}.$$

Decompose B_k and $M = M_R(e)$ relative to $e \oplus e^\perp$ as $B_k = \begin{pmatrix} \beta & b_0^T \\ b_0 & D \end{pmatrix}$, $M = \begin{pmatrix} m & m_0^T \\ m_0 & N \end{pmatrix}$. Then $\beta \leq 1$, $m \leq 2k$, $\text{tr } D \geq H_k - 1$, $\|D\|_2 \leq 1$. Positive block Schur complements give $|b_0^T m_0| \leq \sqrt{\beta m \text{tr}(DN)}$: factor the off-diagonal vectors through the positive square roots and use the operator norm bounded by Frobenius norm. The inequality $2\sqrt{\beta m w} \leq w/2 + 2\beta m$ then yields

$$C_R(e) + 2k \geq \frac{1}{2} \text{tr}(DN).$$

An extra independent feature rotation fixing e leaves its representation and selected indices unchanged, while conjugating N by a Haar rotation on $e^\perp$. Averaging with the preceding reciprocal estimate and (5.3) gives

$$\mathbb{E}_R \frac{1}{C_R(e) + 2k} \leq \frac{800(k-3)}{ck^2(H_k-3)}.$$

Integrate over the independent Gaussian target $B_k^{1/2}z$. For $H_k \geq 6$,

$$\mathbb{E}_R \mathbb{E}_z \frac{kH_k}{C_R(B_k^{1/2}z) + 2k} \leq 1600/c = L/2.$$

Some full-general-position frame has the inner reciprocal expectation less than L , since full general position holds almost surely by nonzero minor polynomials and is preserved by whitening. Markov's inequality gives the asserted bound for every s on that one frame.

For a deterministic choice, enumerate rational full-rank $2k \times k$ matrices by increasing finite numerator and denominator bounds and lexicographic order, and whiten U to $R(U) = U(U^T U)^{-1/2}$. Select the first matrix in full general position whose reciprocal expectation is less than L . Such a rational matrix exists. Averaging supplies a generic frame with expectation less than $3L/4$. Almost every target there meets the interior of a unique supporting simplex facet, so its chosen basis has a strict gap over other representations and varies continuously under small frame perturbations. The reciprocal integrand is bounded by $H_k/2$, so dominated convergence gives continuity of its expectation. Whitening is continuous, genericity is open, and rational matrices are dense. This proves existence of the enumerated choice. Finally $L(s_k + 2k)/(kH_k) \rightarrow 0$ for any $s_k = o(kH_k)$, proving the simultaneous consequence. $\square$

6. A complete coordinate basis in a conference-square frame

The real conference-square construction below is classical; we use the form recorded and proved by Glazyrin [2, Lemma 4.1]. The contribution is the explicit deletion pattern and its complete inverse-cost analysis.

Theorem 6.1. *Let $m \geq 3$ be a prime power with $m \equiv 3 \pmod{4}$, and let χ be the quadratic character on $\mathbb{F}_m$, extended by $\chi(0) = 0$. Put $K_{xy} = \chi(x - y)$, and set*

$$W = -K \otimes K + I_m \otimes J_m - J_m \otimes I_m, \quad S = \begin{pmatrix} 0 & \mathbf{1}^T \\ \mathbf{1} & W \end{pmatrix}, \quad P = (I + S/m)/2.$$

Then P is a real rank- k orthogonal projection, $k = (m^2+1)/2$. There is a deterministically specified k -subset $B \subset \mathbb{F}_m^2$ with

$$\text{tr}(P_{B,B}^{-1}) \leq m^3 + 3m^2 - m - 1 < 4\sqrt{2}k^{3/2}.$$

The added border coordinate is not used.

Proof. Since $\chi(-1) = -1$, $K^T = -K$ and $K\mathbf{1} = 0$. For $h \neq 0$, $\sum_z \chi(z(z-h)) = -1$. After scaling to $h = 1$, count solutions of $v^2 = z(z-1)$: the invertible change $a = 2z - 1 - 2v$, $b = 2z - 1 + 2v$ gives $ab = 1$, with $m-1$ solutions. This proves $KK^T = mI - J$ and $K^2 = J - mI$. The matrix W is symmetric, has diagonal zero and off-diagonal signs, and $W\mathbf{1} = 0$. Its mixed products involving K and J vanish, and

$$W^2 = (J - mI) \otimes (J - mI) + mI \otimes J + mJ \otimes I - 2J \otimes J = m^2I - J.$$

Thus $S^2 = m^2I$, proving that P is a projection of trace k .

Choose a nontrivial additive character ψ and write $u_s(x) = m^{-1/2}\psi(sx)$. These form an orthonormal basis by finite character sums. Convolution gives $Ku_s = \kappa_s u_s$. For $s \neq 0$, $\kappa_s^2 = -m$, and conjugation interchanges the two signs. Define $A_+ = \{s \neq 0 : i\kappa_s = \sqrt{m}\}$, $A_- = -A_+$; each has $(m-1)/2$ elements. Put $d = (m+1)/2$ and $V_+ = (u_s)_{s \in \{0\} \cup A_+}$. Fix a total order of the field, choose the lexicographically first d -row set T maximizing $|\det(V_+)_T|$, and let y_* be its first element. Our actual coordinate set is

$$B = (\mathbb{F}_m \times T) \setminus \{(-y, y) : y \in T \setminus \{y_*\}\}.$$

To analyze it, write $\epsilon_s = 1$ or -1 according as $s \in A_+$ or A_- . An orthonormal complex basis of range P consists of the columns $u_s \otimes u_t$ for $s \neq 0$, $t \in \{0\} \cup A_{\epsilon_s}$, with zero border entry, together with the column having border entry $1/\sqrt{2}$ and every grid entry $1/(\sqrt{2}m)$. Indeed W has eigenvalue $m\epsilon_s\epsilon_t$ on two nonzero frequencies, eigenvalue m on $u_s \otimes u_0$, eigenvalue $-m$ on $u_0 \otimes u_t$, and zero on the constant grid column. There are $(m-1)d+1 = k$ positive-eigenvalue columns. Call their matrix F , so $FF^* = P$. The conjugation-stable range is real, hence a unitary column change produces any real Parseval factor with the same row-submatrix singular values.

Let $V = (V_+)_T$ and $f = \|V^{-1}\|_F^2$. Its determinant is nonzero by full column rank. Replacing a selected row by an outside row changes the determinant by the corresponding entry of $v_x V^{-1}$, so maximality makes each such entry at most one in modulus. Parseval gives

$$f = d + \|(V_+)_T^c V^{-1}\|_F^2 \leq d + d(m-d) = d^2.$$

The restriction of $V_- = (u_s)_{s \in \{0\} \cup A_-}$ to the same T is its complex conjugate, up to column order, and has the same cost.

A range vector with coefficient c_0 in the constant/border column has line sum $\sum_x f(x, y) = c_0/\sqrt{2}$ for every y . The line y_* is fully observed. Thus the missing value at $x = -y$ on every other selected line is recovered from the complete-line sum minus the other observations on that line. After filling the lines, apply the unitary Fourier transform in x . Each nonzero frequency s yields a d -vector across T , equal to $(V_{\epsilon_s})_T$ times its coefficient vector. That square matrix is invertible. The constant coefficient is $\sqrt{2}$ times the full-line sum. This recovers all coefficients, and $|B| = md - (d-1) = k$, so F_B is invertible.

We bound the full inverse cost, including filled-in errors. The squared Frobenius norm of any linear inverse equals the expected squared norm of its output on independent real errors of mean zero and variance one; expansion proves this identity. The recovered constant coefficient contributes $2m$. For nonzero frequency s , errors on an incomplete line have coefficients $m^{-1/2}[\psi(-sx) - \psi(sy)]$, $x \neq -y$, whose squared moduli sum to two. The complete-line Fourier error has variance one and is uncorrelated with its line-sum error. The covariance of the d recovered Fourier measurements is therefore

$$C_s = \text{diag}(2, \dots, 2, 1) + w_s w_s^*, \quad (w_s)_y = \begin{cases} \psi(sy), & y \neq y_*, \\ 0, & y = y_*. \end{cases}$$

Set $V_s = (V_{\epsilon_s})_T$ and $v_s = V_s^{-1}e_*$. Since $s \in A_{\epsilon_s}$, the full phase vector is $\sqrt{m}$ times a column of V_s . Hence

$$V_s^{-1}w_s = \sqrt{m}e_s - \psi(sy_*)v_s, \quad \|V_s^{-1}w_s\|^2 \leq 2m + 2f.$$

The diagonal covariance contributes at most $2f$, so each frequency costs at most $4f + 2m$. Summing marginal squared errors requires no independence across frequencies. Consequently

$$\|F_B^{-1}\|_F^2 \leq 2m + (m-1)(4f + 2m) \leq 2m^2 + 4(m-1)d^2 = m^3 + 3m^2 - m - 1.$$

For $m \geq 3$ this is below $2m^3$, since the difference is $m^2(m-3) + m + 1 > 0$, and $2m^3 < 4\sqrt{2}k^{3/2}$. Finally $P_{B,B} = F_B F_B^*$, so its inverse trace is exactly this cost. $\square$

The prime powers $m = 3^{2j+1}$ give an infinite sequence. The theorem concerns this specified square construction; it does not assert equivalence with every Paley conference matrix of the same order and cannot rule out other difficult conference families.

7. What the obstructions do and do not show

Global determinant-product optimization, balanced determinant-product optimization, mixed-noise minimax, and target-dependent ℓ_1 selection are different operations. The examples establish failure of particular guarantees for each operation, with every maximizing point covered where that quantifier is stated. They do not establish a lower bound for the best possible basis unless a theorem explicitly minimizes over every basis. Conversely the conference theorem supplies a genuine low-cost complete basis, so it rules out that family as a source of a quadratic best-basis lower bound. None of these results determines whether the unrestricted fixed-spectrum Nyström comparison admits a dimension-independent constant.

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